Forbidden heaps and the spectral solution of free-fermion-in-disguise circuits

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Abstract. The spectrum of a free-fermion-in-disguise (FFD) circuit can depend on the order of its gates, even when every local gate is unchanged. We classify the open, one-pass palindromic circuits whose canonical transfer polynomial has commuting coefficients and central inversion. Commuting gate exchanges reduce a permutation to a heap: the relative order of every noncommuting pair. The criterion is local. A heap is admissible precisely when it avoids twenty orientations of a single six-vertex graph, or five patterns up to reflection and reversal. The transfer polynomial follows directly from the circuit expansion, with a binary-nullity grading that counts temporal overlaps rather than selected gates. Its generic spectrum is obtained from polynomial roots, with central-sector sign constraints stated explicitly. A bounded-frontier algorithm computes the required sector polynomials without constructing the many-body matrices. The all-length proof combines a local obstruction, a triangle-interlacement reduction, and exact finite certificates. Complete proofs, pseudocode, and a reproducible verification record are supplied below. The necessity statement concerns this canonical transfer construction, not every conceivable free-fermion representation.

The ordering problem. Free fermions in disguise have a factorized many-body spectrum without an evident Jordan–Wigner reduction of their local spin operators [1, 2]. For quantum circuits, free-fermion solvability is sensitive to the arrangement of the gates. Pozsgay and Fukai posed this ordering problem [3]; Szász-Schagrin, Cristani, Piroli, and Vernier subsequently constructed transfer matrices and polynomial-root solutions for three staircase and brickwork families, conventionally called I, II, and III [4].

Here we treat an arbitrary permutation within a fixed class of palindromes. The result has two parts: a finite forbidden-pattern criterion for the canonical transfer family, and an explicit spectral solution for every accepted heap. Neither translation invariance nor a repeating circuit architecture is required.

The open FFD algebra is generated by Hermitian involutions satisfying

$$h_j^2 = 1, \quad h_j h_k = \begin{cases} -h_k h_j, & 1 \le |j-k| \le 2, \\ h_k h_j, & |j-k| > 2. \end{cases} \tag{1}$$

With $u_j = \exp(i\phi_j h_j/2)$ and a permutation π of $[M]$, set

$$G_\pi = u_{\pi_1} \cdots u_{\pi_M}, \qquad V_\pi = G_\pi G_\pi^T. \tag{2}$$

Here T reverses products and fixes each h_j and the scalar i; it is not the adjoint. Thus the second half retraces the first, with the same angles.

A heap forgets only irrelevant ordering. Draw an edge between labels at distance one or two, obtaining the graph $F_M = P_M^2$. Direct each edge from the gate appearing earlier in the half-word to the gate appearing later. This acyclic orientation is the *heap* D. Its transitive closure is a partial order; its linear extensions are exactly the words related by adjacent exchanges of commuting gates [5]. All those words give the same circuit. Figure 1 illustrates the distinction between the spatial graph and its temporal ordering.

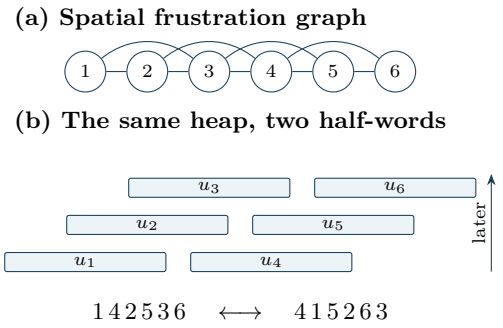


Figure 1. From a word to a heap. (a) An edge means anticommutation, not a physical wire. (b) Draw gate j as an interval centered at j; overlapping intervals cannot pass through one another. Gates in each displayed row commute. Exchanging them changes the word but not the heap or the palindrome. Vertical position shows the relative order in the half-word, not the direction of operator action on a state.

The canonical transfer polynomial. Write $x_j = \cos\phi_j$ and $y_j = \sin\phi_j$. For a selected set S, let $e_D(v, S)$ be the parity of its selected out-neighbors. The set is *feasible* when $e_D(v, S) = 0$ at every selected vertex. Peeling the palindrome from its outside gives the exact expansion

$$V_D = \sum_{S \text{ feasible}} a_D(S) H_D(S), \tag{3}$$

$$a_D(S) = i^{|S|} \prod_{j \in S} y_j \prod_{\substack{j \notin S \\ e_D(j,S)=0}} x_j, \tag{4}$$

where $H_D(S)$ is the product of the selected generators in temporal order.

The nontrivial step is assigning a spectral degree. In temporal coordinates on S, let $N_{ab} = 1$ when $a < b$ and the corresponding generators anticommute. Work over $\mathbb{F}_2$ and define

$$\begin{aligned} K_D(S) &= \operatorname{offdiag}[(I+N)^T(I+N)], \\ \nu_D(S) &= \operatorname{nullity} K_D(S). \end{aligned} \tag{5}$$

A column of $I + N$ marks an event together with its earlier selected neighbors. Thus K records the pairwise overlaps of these *predecessor footprints*, excluding self-overlap. This is not a gate count: a feasible three-gate temporal fork has $\nu = 1$.

The canonical polynomial is

$$\boxed{\mathsf{A}_D(z) = \sum_{S \text{ feasible}} a_D(S) z^{\nu_D(S)} H_D(S).} \tag{6}$$

It obeys $\mathsf{A}_D(1) = V_D$, $\mathsf{A}_D(0) = a_0\mathbf{1}$ with $a_0 = \prod_j x_j$, and $\mathsf{A}_D(z)^\dagger = \mathsf{A}_D(-z)$ for real z.

The local classification. The obstruction graph H has two triangles sharing an edge and a pendant edge at each end:

$$E(H) = \{AB, BC, BD, CD, CE, DE, EF\}. \tag{7}$$

Its relevant copies occupy

$$(A, B, C, D, E, F) = (a, a+2, a+3, a+4, a+5, a+7). \tag{8}$$

Orient the edges by any one of the five words

$$\begin{gathered}\texttt{ADBCFE}, \quad \texttt{ADBCEF}, \quad \texttt{ACBFED}, \\ \texttt{ACBDFE}, \quad \texttt{ACBDEF}.\end{gathered} \tag{9}$$

Reversing every arrow and reflecting $A \leftrightarrow F$, $B \leftrightarrow E$, $C \leftrightarrow D$ produces twenty distinct forbidden orientations (Fig. 2). The other two sites in the eight-site window are ignored, not deleted from the circuit.

Theorem 1 (Canonical FFD classification)**.** *For every open-chain length and every heap D, the following are equivalent: (i) no translated copy in (8) has a forbidden orientation; (ii) the family (6) satisfies*

$$\deg \mathsf{A}_D \le s_M := \lceil M/3 \rceil, \tag{10}$$

$$[\mathsf{A}_D(z), \mathsf{A}_D(w)] = 0, \tag{11}$$

$$\mathsf{A}_D(z)\mathsf{A}_D(-z) \in Z(\mathcal{A}_M)[z^2]. \tag{12}$$

These are identities for arbitrary angles. Sufficiency is proved by exact finite certificates valid at every length.

The criterion takes $O(M)$ comparisons once the positions of the labels in π are known. All heaps of lengths $M \le 7$ pass. At $M = 8$, precisely 1,054 of the 1,458 heaps pass; these are counts of heaps, not permutations. A forbidden heap fails the canonical commutation identity generically, although special angle choices can remove the obstruction. Crucially, this does not exclude a different transfer family or an exceptional free spectrum.

Why the criterion works. Necessity is genuinely local. Turn off all gates outside a forbidden copy by setting their angles to zero. In its six-generator algebra, the full-support coefficient of $[Q_1, Q_3]$, where $\mathsf{A}(z) = \sum_r z^r Q_r$, is nonzero. For the five representatives it is, after division by $\prod_{X=A}^{F} y_X$, respectively

$$-2x_E, \quad -2x_F, \quad 2x_D, \quad 2x_E, \quad 2x_F. \tag{13}$$

Sufficiency has a geometric reduction. The nullity is the number of connected components of $F_M[S]$ plus the nullity of an auxiliary graph whose vertices are the selected consecutive triangles. Two such triangles are linked according to the temporal order of their middle events. Avoidance forces this auxiliary graph to be a union of paths. A path contributes one to the nullity precisely when it contains an odd number of vertices. Consequently

$$\nu_D(S) = c(F_M[S]) + \#\{\text{odd interlace paths}\}. \tag{14}$$

Only a fixed frontier must be remembered as the chain is scanned; a 32-state transducer evaluates the grade.

Every angle coefficient of the commutator and of the noncentral inversion product then becomes a signed weighted sum recognized by a finite automaton. Exact quotients have 9,756 and 7,195 states. Their vanishing is certified by invariant subspaces of dimensions 590 and 427 over $\mathbb{Q}(z, w)$ and $\mathbb{Q}(z)$, respectively. The full verification checks all $17{,}700 + 12{,}810$ transition images. This is an all-length proof, not an extrapolation from short chains. The supplementary material gives the reductions, the finite-state definitions, and executable pseudocode.

From the transfer polynomial to the spectrum. Fix a central sector χ. Central inversion becomes

$$\mathsf{A}_{D,\chi}(z)\mathsf{A}_{D,\chi}(-z) = p_{D,\chi}(z)\mathbf{1}. \tag{15}$$

For generic angles, $p_{D,\chi}$ has s_M distinct nonzero root pairs $\{r_{\chi,k}, -r_{\chi,k}\}$. Simultaneous diagonalization and (15) give

$$\boxed{v_{\chi,\boldsymbol{\sigma}} = a_0 \prod_{k=1}^{s_M} \left(1 + \frac{\sigma_k}{r_{\chi,k}}\right), \qquad \boldsymbol{\sigma} \in \mathcal{C}_\chi.} \tag{16}$$

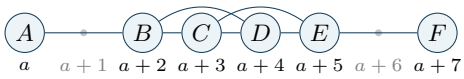
The admissible signs form an affine binary code with equal multiplicities. In a fixed irreducible sector they are unrestricted unless $M \equiv 1 \pmod 3$; in that case one product-of-signs constraint remains. Its value is obtained from the highest transfer coefficient, as explained in Sec. S15.

For real angles on the generic locus, choose $r_{\chi,k} = iu_{\chi,k}$ with $u_{\chi,k} > 0$. Changing the kth sign from -1 to $+1$ multiplies the eigenvalue by

$$q_{\chi,k} = \frac{iu_{\chi,k} + 1}{iu_{\chi,k} - 1} = e^{-i\omega_{\chi,k}}, \quad \omega_{\chi,k} = 2 \arctan \frac{1}{u_{\chi,k}}. \tag{17}$$

A fixed-sector parity constraint may require signs to change in pairs. The displayed quantities are the elementary factors, not a claim that every individual flip is allowed in each sector.

(a) One six-site test in an eight-site window



Compare only the seven displayed edges.
The two gray sites are spectators in this test.

(b) Five representatives; reflection and reversal generate all twenty

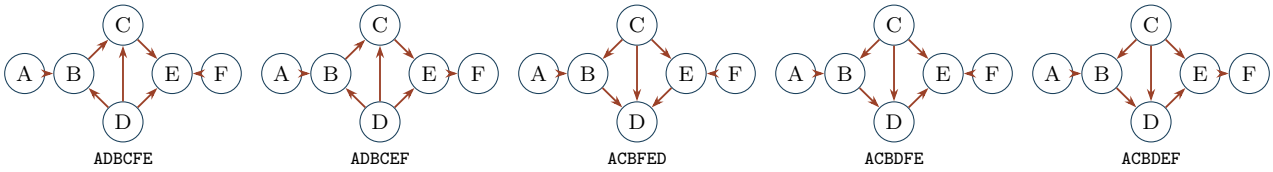


Figure 2. The forbidden heaps. A representative word specifies only relative orders on the seven edges of H; adjacent commuting letters may be exchanged freely. Each panel in (b) represents four distinct orientations under temporal reversal and spatial reflection. Applying this test at every position gives the exact canonical classification.

Computing the polynomial, not a large matrix. The identity coefficient of the inversion product is the positive sum

$$p_0(z) = \sum_{S \text{ feasible}} z^{2\nu_D(S)} \prod_{j\in S} y_j^2 \prod_{\substack{j\notin S \\ e_D(j,S)=0}} x_j^2. \qquad (18)$$

When inversion is scalar in the full algebra, $p_0 = p_{D,\chi}$ for every sector. *Centrality alone is not enough to make this replacement.* In general one also computes the coefficients supported on $\ker_{\mathbb{F}_2} A_M$, where A_M is the frustration adjacency matrix, and evaluates them in χ. This kernel has dimension at most two. A paired version of the same frontier scan computes all sector polynomials in $O(Ms_M)$ arithmetic operations, with a size-independent state bound (Sec. S14). Thus the general root prescription is constructive even when the center is nontrivial.

The familiar half-words I, II, and III are

$$\begin{aligned} &\mathrm{I} : (1, 2, \ldots, M), \\ &\mathrm{II} : (2, 4, \ldots, M; 1, 3, \ldots, M-1), \\ &\mathrm{III} : (3, 6, \ldots, M; 2, 5, \ldots, M-1; 1, 4, \ldots, M-2), \end{aligned} \qquad (19)$$

with the usual even-M and $3 \mid M$ conventions in II and III. They pass the local test. These are the regular families solved in Ref. [4]; the present construction also covers heaps without a repeating architecture.

As a nonperiodic example, take $M = 9$, $\pi = (9, 4, 2, 5, 7, 1, 3, 8, 6)$, and $\phi_j = 0.17 + 0.029(j-1) + 0.004 \sin j$. Direct evaluation verifies scalar inversion here. The normalized mode polynomial $P(x) = p_0(i\sqrt{x})/p_0(0)$ is

$$\begin{aligned} P(x) = 1 &- 1.02263185775715x \\ &+ 0.209346393878477x^2 \\ &- 0.00725922352756874x^3. \end{aligned} \qquad (20)$$

The roots are approximately 1.31645, 4.55642, and 22.96581, with occupation-flip phases 1.433758, 0.876225, and 0.411435. Using unrounded coefficients, the eight eigenvalues from (16), each with multiplicity 64, agree with an independent 512×512 spin-circuit calculation to 1.2×10^{-14}.

Scope. The result resolves the ordering problem for the canonical nullity-graded transfer construction: acceptance is both necessary and sufficient, and supplies a polynomial-root solution. It does not turn failure of that particular construction into a no-go theorem for every other definition of free-fermion solvability. Repeated roots, vanishing cosines, and other nongeneric points are obtained by limits of the unnormalized polynomial identities; normalized root formulas should not be used there without taking those limits.

Reproducibility. The accompanying single-file code archive contains the complete proof sources and certificates, a sector-aware spectral solver, and independent checks. Both formal automata were regenerated byte for byte, all characteristic-zero certificate identities were rerun, and the independent finite audits were repeated. No finite-field specialization is used as the final proof.

References

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[2] S. J. Elman, A. Chapman, and S. T. Flammia, Free fermions behind the disguise, Commun. Math. Phys. **388**, 969–1003 (2021), arXiv:2012.07857.

[3] K. Fukai and B. Pozsgay, Quantum circuits with free fermions in disguise, J. Phys. A: Math. Theor. **58**, 175202 (2025), arXiv:2402.02984.

[4] D. Szász-Schagrin, D. Cristani, L. Piroli, and E. Vernier, Construction and simulability of quantum circuits with free fermions in disguise, Quantum Sci. Technol. **11**, 015044 (2026), arXiv:2509.22585.

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Supplementary material

Forbidden heaps and the spectral solution of free-fermion-in-disguise circuits

This supplement contains the complete algebraic reductions, the finite exact certificate argument, and the constructive sector-by-sector spectral solution. The long-chain cancellations use the finite exact certificates included in the archive. The final sections document the independently rerun checks and the accompanying code archive.

S1 Open FFD circuits and heaps

S1.1 The algebra

Fix $M \geq 1$. The open FFD $*$-algebra $\mathcal{A}_M$ is generated by Hermitian involutions $h_1, \ldots, h_M$ satisfying

$$h_j^2 = \mathbf{1}, \qquad \{h_j, h_k\} = 0 \quad (1 \leq |j-k| \leq 2), \qquad [h_j, h_k] = 0 \quad (|j-k| > 2). \tag{S1}$$

The frustration graph is

$$F_M = P_M^2, \qquad \{j,k\} \in E(F_M) \Longleftrightarrow 1 \leq |j-k| \leq 2.$$

The open chain and its distance-one and distance-two frustration edges are illustrated in figure S1.

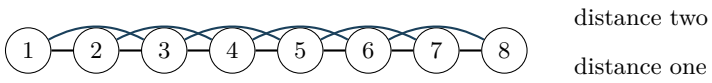


Figure S1. The frustration graph $F_M = P_M^2$. Generators anticommute along both kinds of edge and commute otherwise.

Choose a spatial normal order $h^x = h_1^{x_1} \cdots h_M^{x_M}$ for $x \in \mathbb{F}_2^M$. If A_M is the adjacency matrix of F_M over $\mathbb{F}_2$, then

$$h^x h^y = (-1)^{x^{\mathsf{T}} A_M y} h^y h^x. \tag{S2}$$

Hence h^x is central exactly when $A_M x = 0$. We write $\mathcal{Z}_M = Z(\mathcal{A}_M)$.

S1.2 Palindromic permutation circuits

For real angles ϕ_j, define

$$u_j = \exp\left(\frac{i\phi_j}{2} h_j\right) = \cos\left(\frac{\phi_j}{2}\right)\mathbf{1} + i \sin\left(\frac{\phi_j}{2}\right) h_j. \tag{S3}$$

Let T be the complex-linear anti-involution determined by $h_j^T = h_j$ and $(XY)^T = Y^T X^T$. For a permutation $\pi = (\pi_1, \ldots, \pi_M)$, set

$$G_\pi = u_{\pi_1} \cdots u_{\pi_M}, \qquad V_\pi = G_\pi G_\pi^T. \tag{S4}$$

Thus the second half retraces the first without complex conjugating the scalar i. The circuit is a one-pass palindrome: every generator occurs once in the half-word.

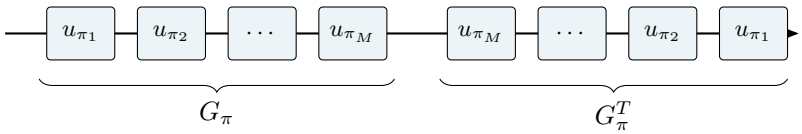


Figure S2. The palindromic circuit $V_\pi = G_\pi G_\pi^T$. Only the relative order of noncommuting gates matters.

S1.3 Heaps: order modulo commuting swaps

Orient every edge $j - k$ of F_M from the label that occurs earlier in π to the label that occurs later. The result is an acyclic orientation D_π. Two permutations determine the same orientation exactly when they differ by adjacent exchanges of commuting labels. Since those gates commute, the circuit depends only on the orientation. We call it the *heap* D, following the trace-monoid viewpoint [5].

We write $v \to w$ when the edge vw is directed from v to w. The directed adjacency matrix is

$$(E_D)_{vw} = \mathbf{1}_{\{v\to w\}}. \tag{S5}$$

Then $E_D + E_D^{\mathsf{T}} = A_M$.

A convenient insertion encoding will be used by the automata. After the first edge is oriented, a new site j belongs to the triangle $j-2, j-1, j$. Transitivity allows exactly three temporal positions: before both old sites, between them, or after both. We denote this ternary choice by $q_j \in \{0,1,2\}$. Consequently the number of acyclic orientations is $2 \cdot 3^{M-2}$ for $M \geq 2$.

S2 Exact expansion of the palindrome

Put

$$x_j = \cos\phi_j, \qquad y_j = \sin\phi_j. \tag{S6}$$

The doubled gate is $u_j^2 = x_j + iy_jh_j$. The exact expansion is most transparent if the palindrome is peeled from its outside.

For $S \subseteq [M]$, define the selected outdegree parity

$$e_D(v,S) = \#\{w \in S : v \to w\} \pmod 2. \tag{S7}$$

Definition S1 (Feasible support)**.** *A support $S \subseteq [M]$ is D-feasible if*

$$e_D(v,S) = 0 \qquad (v \in S). \tag{S8}$$

We denote the family by $\mathcal{F}(D)$.

For $S \in \mathcal{F}(D)$, let $H_D(S)$ be the product of the selected generators in any linear extension of $D|_S$. This is well defined because incomparable vertices are nonadjacent in F_M and hence commute.

Proposition S2 (Exact feasible-support expansion)**.** *The palindrome associated with a heap D is*

$$V_D = \sum_{S\in\mathcal{F}(D)} a_D(S)H_D(S), \tag{S9}$$

where

$$a_D(S) = i^{|S|} \prod_{v\in S} y_v \prod_{\substack{v\notin S \\ e_D(v,S)=0}} x_v. \tag{S10}$$

Proof. Let P be a graph-Clifford monomial in the generators that lie inside the current pair of gates. Direct multiplication gives

$$u_vPu_v = \begin{cases} (x_v + iy_vh_v)P, & [h_v,P]=0, \\ P, & \{h_v,P\}=0. \end{cases} \tag{S11}$$

The commutation parity of h_v with the later selected monomial is exactly $e_D(v,S)$. If it is odd, the two scalar terms and the two h_v-terms cancel pairwise, leaving P; hence v must be absent. If it is even, one may omit v with factor x_v, or select it with factor iy_v. A selected vertex is therefore allowed exactly under (S8). Multiplying the local factors yields (S10). □

Remark S3 (Why feasibility is temporal)**.** *The condition is not merely even degree in the induced frustration graph. Only later selected neighbors enter. It therefore depends on the heap orientation. This asymmetry is what permits nonstable feasible supports and is the source of the nontrivial grading below.*

S3 The temporal-footprint grading

The correct spectral degree is not $|S|$. Folded FFD circuits contain composite monomials, including trilinear terms, that carry degree one. The nullity below supplies a grading compatible with the temporal overlap of the selected gates.

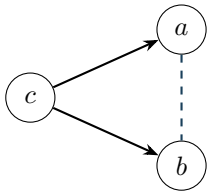


Figure S3. A temporal fork. The two arrows contribute $N_{ca}N_{cb} = 1$, so the common-predecessor term toggles the mutual relation of a and b.

S3.1 Temporal adjacency and predecessor footprints

Let $S \in \mathcal{F}(D)$, and list its elements in temporal order,

$$S = \{s_1 \prec s_2 \prec \cdots \prec s_n\}.$$

Define the strictly upper-triangular matrix $N = N_{D,S} \in M_n(\mathbb{F}_2)$ by

$$N_{ab} = \mathbf{1}_{\{a<b,\ 1\le|s_a-s_b|\le 2\}}. \tag{S12}$$

Thus row a records the later selected generators that anticommute with h_{s_a}, and feasibility is

$$N\mathbf{1} = 0. \tag{S13}$$

For the later event s_a, define its closed predecessor footprint

$$p_a = \mathbf{e}_a + \sum_{c<a} N_{ca}\mathbf{e}_c. \tag{S14}$$

The footprint marks the event itself and every earlier selected event whose commutation parity is flipped by selecting s_a. If the footprints are placed as columns, then

$$B = (p_1 \ \cdots \ p_n) = I + N. \tag{S15}$$

Their binary Gram matrix is

$$G = B^{\mathsf{T}}B = I + N + N^{\mathsf{T}} + N^{\mathsf{T}}N. \tag{S16}$$

The diagonal of G is unavoidable self-overlap. Since B is unitriangular, G is always nonsingular and its full nullity is trivial. The mutual overlaps are captured by deleting the diagonal.

Definition S4 (Canonical grading matrix)**.** *Set*

$$K_D(S) = \operatorname{offdiag}\big[(I + N_{D,S})^{\mathsf{T}}(I + N_{D,S})\big] = N + N^{\mathsf{T}} + \operatorname{offdiag}(N^{\mathsf{T}}N), \tag{S17}$$

and define

$$\nu_D(S) = \operatorname{nullity}_{\mathbb{F}_2} K_D(S). \tag{S18}$$

For $a \neq b$, the entry has the explicit form

$$(K_D(S))_{ab} = \mathbf{1}_{\{s_a\sim s_b\}} + \#\{c : s_c \prec s_a, s_b,\ s_c \sim s_a, s_b\} \pmod 2. \tag{S19}$$

The first term is direct anticommutation. The second counts common temporal predecessors: two later selected events overlap when they flip the same earlier parity register.

Example S5 (One temporal fork)**.** *For*

$$N = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

the direct frustration graph is a two-leaf star. The common predecessor adds the leaf-to-leaf edge, and

$$K = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

has binary nullity one. Thus this feasible trilinear monomial has spectral degree one.

S3.2 Elementary properties

Lemma S6 (Basic grading properties)**.** *For every feasible support S:*

(a) *$\nu_D(S)$ is independent of the chosen linear extension;*

(b) *$K_D(S)\mathbf{1} = 0$;*

(c) *$\nu_D(S) \equiv |S| \pmod 2$;*

(d) *$H_D(S)$ is Hermitian.*

Proof. Adjacent exchanges between linear extensions involve incomparable, hence nonadjacent, vertices. They only permute matching rows and columns of $K_D(S)$, proving (a).

For (b), feasibility gives $N\mathbf{1} = 0$. Over $\mathbb{F}_2$,

$$\operatorname{diag}(N^{\mathsf{T}}N) = N^{\mathsf{T}}\mathbf{1}, \qquad \operatorname{offdiag}(N^{\mathsf{T}}N)\mathbf{1} = N^{\mathsf{T}}\mathbf{1}.$$

Therefore

$$K_D(S)\mathbf{1} = N\mathbf{1} + N^{\mathsf{T}}\mathbf{1} + N^{\mathsf{T}}\mathbf{1} = 0.$$

The matrix $K_D(S)$ is symmetric with zero diagonal, hence alternating over $\mathbb{F}_2$. Its rank is even, which proves (c).

Reversing $H_D(S)$ introduces one minus sign for every edge of $F_M[S]$. The number of such edges is the sum of the selected outdegrees in $D[S]$, which is even by feasibility. Hence the monomial is Hermitian, proving (d). □

The identity $K_D(S)\mathbf{1} = 0$ is also a useful retrospective explanation for the correction $N^{\mathsf{T}}N$. The direct frustration matrix $N + N^{\mathsf{T}}$ need not be singular on a feasible support. The common-predecessor term cancels its residual column-parity vector exactly.

S3.3 The canonical Baxter family

Definition S7 (Canonical family)**.** *Define*

$$\boxed{\mathsf{A}_D(z) = \sum_{S \in \mathcal{F}(D)} a_D(S) z^{\nu_D(S)} H_D(S) = \sum_{r \geq 0} z^r Q_{D,r}.} \tag{S20}$$

By construction,

$$\mathsf{A}_D(1) = V_D, \qquad \mathsf{A}_D(0) = a_0\mathbf{1}, \qquad a_0 = \prod_{j=1}^{M} x_j. \tag{S21}$$

The last identity uses $K_D(S)\mathbf{1} = 0$: every nonempty feasible support has positive nullity.

Corollary S8 (Adjoint parity)**.** *For every r,*

$$Q_{D,r}^{\dagger} = (-1)^r Q_{D,r}. \tag{S22}$$

Consequently $\mathsf{A}_D(z)^{\dagger} = \mathsf{A}_D(-z)$ for real z.

Proof. The only nonreal factor in $a_D(S)$ is $i^{|S|}$. By lemma S6, $|S| \equiv \nu_D(S) \pmod 2$, while $H_D(S)$ is Hermitian. Group terms of fixed nullity. □

S4 Forbidden heaps and the Baxter property

S4.1 The six-vertex obstruction

Let H have vertices A, B, C, D, E, F and edges

$$E(H) = \{AB, BC, BD, CD, CE, DE, EF\}. \tag{S23}$$

It consists of two triangles BCD and CDE sharing the edge CD, with pendant edges AB and EF.

For a word containing the six letters once, orient every edge from the earlier letter to the later one. Let τ reverse the word, and let ρ be the spatial reflection

$$A \leftrightarrow F, \qquad B \leftrightarrow E, \qquad C \leftrightarrow D.$$

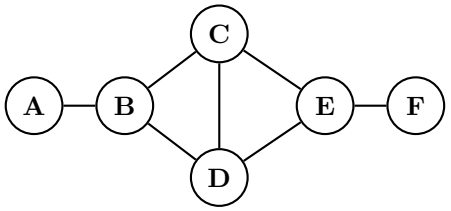


Figure S4. The obstruction graph H. Twenty of its seventy-two acyclic orientations are forbidden.

Start from the five temporal orders

$$ADBCFE, \qquad ADBCEF, \qquad ACBFED, \qquad ACBDFE, \qquad ACBDEF, \tag{S24}$$

where spaces are inserted only for readability. Applying $\tau^\epsilon \rho^\delta$, $\epsilon, \delta \in \{0,1\}$, gives twenty distinct orientations. Their set is denoted $\mathfrak{B}$. A complete seven-bit signature list is given in section S16.

For $1 \le a \le M-7$, the sites

$$A = a, \quad B = a+2, \quad C = a+3, \quad D = a+4, \quad E = a+5, \quad F = a+7 \tag{S25}$$

induce a copy $H_a \subset P_M^2$.

Definition S9 (H-avoidance)**.** *A heap D is H-avoiding if $D|_{H_a} \notin \mathfrak{B}$ for every allowed translation a.*

S4.2 Canonical Baxter property

Let

$$s_M = \alpha(P_M^2) = \left\lceil \frac{M}{3} \right\rceil. \tag{S26}$$

All angle identities are understood in the quotient ring

$$\mathscr{R}_M = \mathbb{C}[x_1, y_1, \ldots, x_M, y_M]/(x_j^2 + y_j^2 - 1 : 1 \le j \le M). \tag{S27}$$

Definition S10 (Canonical Baxter property)**.** *A heap D has the canonical Baxter property if*

$$\deg_z \mathsf{A}_D(z) \le s_M, \tag{S28}$$

$$[\mathsf{A}_D(z), \mathsf{A}_D(w)] = 0, \tag{S29}$$

$$\mathsf{A}_D(z)\mathsf{A}_D(-z) \in \mathscr{Z}_M[z^2]. \tag{S30}$$

Theorem 1 says that this property is equivalent to avoiding the twenty local orientations.

S5 Physical consequence: a product spectrum

The algebraic theorem immediately implies the usual free-fermion spectral form. We include the derivation because it explains why the three Baxter conditions are the right target.

An *affine sign code* $\mathcal{C} \subseteq \{\pm 1\}^s$ is a coset of a subgroup under componentwise multiplication; equivalently, it is defined by parity equations

$$\prod_{k=1}^{s} \sigma_k^{c_{\ell k}} = \zeta_\ell, \qquad c_{\ell k} \in \mathbb{F}_2, \quad \zeta_\ell \in \{\pm 1\}.$$

Theorem S11 (Baxter identities and the FFD spectrum)**.** *Suppose D has the canonical Baxter property. On a Zariski-open set of angle parameters, the following holds in every irreducible central sector χ. The charges have a common, z-independent eigenspace decomposition, and their joint eigenvalue polynomials are*

$$\Lambda_{\chi,\boldsymbol{\sigma}}(z) = a_0 \prod_{k=1}^{s_M} (1 + \sigma_k \lambda_{\chi,k} z), \qquad \boldsymbol{\sigma} \in \mathcal{C}_\chi, \tag{S31}$$

where the nonzero $\lambda_{\chi,k}$ are distinct up to sign, $\mathcal{C}_\chi$ is an affine sign code, and every allowed sign vector occurs with the same multiplicity. Conversely, a common eigenbasis with the product form (S31) implies the canonical Baxter property.

Proof. By corollary S8, the operators $i^{-r}Q_{D,r}$ are Hermitian. Commutation therefore gives a common eigenbasis in each central sector. Let $\Lambda_a(z)$ be one joint eigenvalue polynomial. The degree and constant term obey

$$\deg \Lambda_a \le s_M, \qquad \Lambda_a(0) = a_0.$$

Central inversion becomes a scalar identity

$$\Lambda_a(z)\Lambda_a(-z) = P_\chi(z^2), \tag{S32}$$

with the same polynomial P_χ for every joint eigenspace in the sector.

To see the generic root structure, specialize all angles outside the maximal stable set

$$I = \{1, 4, 7, \ldots\} \cap [M], \qquad |I| = s_M,$$

to $y_j = 0, x_j = 1$. Every surviving support lies in I, so $N = K = 0$ and $\nu_D(S) = |S|$. Hence, independently of the heap,

$$\mathsf{A}_D(z) = \prod_{j\in I}(x_j + izy_j h_j), \tag{S33}$$

and

$$P_\chi(z^2) = \prod_{j\in I}(x_j^2 + y_j^2 z^2).$$

At generic values this polynomial has s_M distinct nonzero root pairs. Therefore, away from a proper algebraic subset,

$$P_\chi(z^2) = a_0^2 \prod_{k=1}^{s_M}(1 - \lambda_{\chi,k}^2 z^2). \tag{S34}$$

A polynomial Λ_a satisfying (S32) contains exactly one zero from every pair $\{\lambda_{\chi,k}^{-1}, -\lambda_{\chi,k}^{-1}\}$. It cannot contain both, because that would make a double zero in the product, and it cannot contain neither. This proves (S31) with some sign vector.

At the decoupled point, the commuting generators h_j, $j \in I$, span an isotropic subspace of the graph-Clifford commutation form. In a fixed central sector, their simultaneous characters are exactly those whose restriction to the central intersection agrees with χ; this is an affine sign code. Any two admissible characters are related by conjugation with a graph-Clifford monomial, so their multiplicities agree. Analytic continuation on the generic locus can only permute root pairs and exchange the two roots inside a pair, which acts by a signed permutation and preserves the affine-code property.

Conversely, a common product eigenbasis makes all charges commute. The product with $z \mapsto -z$ is scalar in every irreducible representation, hence central in the finite-dimensional semisimple algebra. The degree bound is immediate. □

The constructive sector polynomials, the exact sign constraints, and the reality of the generic physical root pairs are developed in Secs. S14 and S15. In particular, central inversion must not be silently replaced by scalar inversion in the full algebra.

S6 Necessity: the local forbidden obstruction

The reverse implication in theorem 1 is local and exact.

Theorem S12 (Six-site obstruction)**.** *Among the seventy-two acyclic orientations of H, the twenty orientations in $\mathfrak{B}$ are exactly those for which the coefficient of the full-support monomial $h_A h_B h_C h_D h_E h_F$ in $[Q_{D,1}, Q_{D,3}]$ is nonzero. For the five representatives (S24), after factoring $\prod_{X=A}^{F} y_X$, the coefficients are*

$$-2x_E, \qquad -2x_F, \qquad 2x_D, \qquad 2x_E, \qquad 2x_F. \tag{S35}$$

Proof. There are only seventy-two orientations and sixty-four supports. The supplied exact verifier enumerates the feasible supports, constructs $K_D(S)$ by binary arithmetic, retains the degree-one and degree-three terms, and multiplies graph-Clifford monomials in a fixed spatial normal order. It finds nonzero full-support coefficients in exactly the twenty stated orientations.

The calculation can be inspected by hand in each representative. For the temporal order $ACBDEF$, for instance,

$$Q_1 \supset ix_F y_B y_C y_E\, h_B h_C h_E, \qquad Q_3 \supset -iy_A y_D y_F\, h_A h_D h_F.$$

The supports are complementary and the monomials anticommute. Their commutator contributes

$$2x_F \prod_{X=A}^{F} y_X\, h_A h_B h_C h_D h_E h_F,$$

and there is no second pair with the same full support. The remaining representatives are equally short. Temporal reversal and spatial reflection give the other fifteen forbidden orientations. □

Corollary S13 (Forbidden copy implies failure)**.** *If a heap D contains a translated copy H_a with orientation in $\mathfrak{B}$, then D does not have the canonical Baxter property.*

Proof. Specialize all angles outside the six sites to $x_j = 1, y_j = 0$. The corresponding gates become the identity, and the six-site coefficient from theorem S12 survives. Hence $[Q_{D,1}, Q_{D,3}]$ is not the zero polynomial. □

S7 An all-heap degree bound

The degree estimate does not require H-avoidance.

Theorem S14 (Sink-nullity bound)**.** *For every heap D and every feasible support S,*

$$\nu_D(S) \le |\operatorname{Sink}(D[S])|. \tag{S36}$$

Consequently

$$\deg_z \mathsf{A}_D(z) \le \left\lceil \frac{M}{3} \right\rceil \tag{S37}$$

for every heap.

The proof establishes injectivity of the restriction map

$$\ker K_D(S) \longrightarrow \mathbb{F}_2^{\operatorname{Sink}(D[S])}. \tag{S38}$$

Suppose $q \in \ker K_D(S)$ and $q_v = 0$ at every sink. We must prove $q = 0$. By (S19), the matrix has spatial bandwidth four: $(K_D(S))_{ij} = 0$ whenever $|i - j| > 4$. This permits a left-to-right scan that retains only four sites.

S7.1 Local scan

A boundary state records, for each retained site,

$$(\text{real}, \text{selected}, q_v, \text{current residual of } (Kq)_v),$$

together with the orientations of distance-one and distance-two edges in the window. On appending a new site t:

1. insert the direct P_M^2 contributions between t and $t-1, t-2$;
2. finalize the selected out-neighborhood of $k = t-2$, whose possible neighbors all lie in $\{k-2, k-1, k+1, k+2\}$;
3. enforce feasibility at k;
4. if k has no selected out-neighbor, enforce the sink condition $q_k = 0$;
5. toggle the clique on the selected out-neighbors of k, which inserts the common-predecessor contribution to K;
6. check the now-complete equation $(Kq)_{t-4} = 0$ and forget site $t-4$.

Acyclicity is local because P_M^2 is chordal: it is enough to require every consecutive triangle to be transitively oriented.

Algorithm S1 Four-site sink-nullity scan

```
S ← {four blank boundary sites}
repeat
    S_new ← S
    for all states σ ∈ S do
        for all legal choices for the next site and incident orientations do
            finalize the site two steps behind the new site
            reject if transitivity, feasibility, or a sink constraint fails
            add direct and common-predecessor contributions to the residuals
            reject if the oldest completed row of Kq = 0 fails
            shift the four-site window and insert the resulting state
    S ← S_new
until no new state is found
for all σ ∈ S do
    append four blank sites and test whether σ flushes legally
reject the certificate if a flushable state has ever seen q_v = 1
```

S7.2 Finite closure certificate

The exact state counts after one, two, three, and four real sites are

$$3, \qquad 18, \qquad 96, \qquad 504.$$

Closing the transition relation under arbitrary additional sites stabilizes at

$$908 \tag{S39}$$

reachable states. Exactly 191 of them can be flushed through the right boundary. None of these flushable states has ever encountered $q_v = 1$. Therefore a kernel vector vanishing on all sinks is zero, proving (S36).

Finally, two adjacent selected vertices cannot both be sinks. Thus the sinks form a stable set of P_M^2, and

$$|\operatorname{Sink}(D[S])| \le \alpha(P_M^2) = \left\lceil \frac{M}{3} \right\rceil .$$

This proves (S37).

Remark S15 (Nature of the certificate)**.** *The state set is not a bounded-size check in M; it is a closed transition system. Once closure is reached, every longer scan remains inside the same finite set. The certificate therefore proves the inequality at every length.*

S8 A triangle-interlace formula for the grading

The grading matrix $K_D(S)$ is defined in temporal coordinates. For the finite-state proof it is useful to replace it by a spatially banded matrix on consecutive triangles.

Let $S \in \mathcal{F}(D)$ and put

$$G_S = P_M^2[S].$$

For every i such that $i, i+1, i+2 \in S$, let τ_i be the three-edge cycle on these vertices. Write

$$T(S) = \{i : i, i+1, i+2 \in S\}. \tag{S40}$$

The oriented triangle is transitive. Its *temporal median* $m_i \in \{i, i+1, i+2\}$ is the unique vertex with one incoming and one outgoing triangle edge.

Define the alternating matrix $\Omega_D[T(S)]$, indexed by $T(S)$, by

$$\Omega_{i,i+1} = 1 + \mathbf{1}_{\{m_i = m_{i+1}\}}, \tag{S41}$$

$$\Omega_{i,i+2} = \mathbf{1}_{\{m_i = m_{i+2}\}}, \tag{S42}$$

and $\Omega_{ij} = 0$ for $|i - j| > 2$.

Theorem S16 (Triangle-interlace formula)**.** *For every heap D and every feasible support S,*

$$\boxed{\nu_D(S) = c(G_S) + \operatorname{nullity}_{\mathbb{F}_2} \Omega_D[T(S)],} \tag{S43}$$

where $c(G_S)$ is the number of connected components, with $c(\varnothing) = 0$.

Proof. Let B be the unoriented vertex-edge incidence matrix of G_S over $\mathbb{F}_2$. Let P be the tail-incidence matrix,

$$P_{v,e} = 1 \iff v \text{ is the tail of the oriented edge } e.$$

Feasibility makes the diagonal vanish in the identity

$$N_{D,S} = PB^{\mathsf{T}}. \tag{S44}$$

Indeed, off the diagonal the right-hand side records the orientation, while its diagonal records selected outdegree parity.

The grading matrix can be written

$$K_D(S) = BB^{\mathsf{T}} + N_{D,S}^{\mathsf{T}} N_{D,S}. \tag{S45}$$

Off the diagonal this is the direct adjacency plus common-predecessor parity. On the diagonal, BB^{T} contributes degree parity and $N^{\mathsf{T}}N$ contributes indegree parity; their sum is outdegree parity, which vanishes. Substituting (S44) gives

$$K_D(S) = BJB^{\mathsf{T}}, \qquad J = I + P^{\mathsf{T}}P. \tag{S46}$$

For a fixed tail, the outgoing edges form one block of J, and that block is the adjacency matrix of a complete graph on an even number of vertices. Over $\mathbb{F}_2$, the adjacency matrix of an even complete graph squares to the identity. Hence

$$J^2 = I. \tag{S47}$$

Let $Z = \ker B$ be the cycle space of G_S. Since $\operatorname{im} B^{\mathsf{T}} = Z^{\perp}$, multiplication by B^{T} gives an exact sequence

$$0 \longrightarrow \ker B^{\mathsf{T}} \longrightarrow \ker(BJB^{\mathsf{T}}) \longrightarrow Z^{\perp} \cap JZ \longrightarrow 0. \tag{S48}$$

The first term has dimension $c(G_S)$. Because J is involutory, multiplication by J identifies the intersection with the radical of the bilinear form

$$\beta(z, z') = z^{\mathsf{T}} J z' \tag{S49}$$

restricted to Z. Therefore

$$\nu_D(S) = c(G_S) + \dim \operatorname{rad}(\beta|_Z). \tag{S50}$$

Every induced subgraph of P_M^2 is chordal and has clique number at most three. Its cycle space is generated by consecutive triangles. These cycles are independent because the distance-two edge $\{i, i+2\}$ occurs only in τ_i. Thus $\{\tau_i : i \in T(S)\}$ is a basis of Z.

In this basis,

$$\beta(\tau_i, \tau_j) = \tau_i^{\mathsf{T}} \tau_j + (P\tau_i)^{\mathsf{T}} (P\tau_j).$$

Inside a transitively oriented triangle, the parity of outgoing triangle edges equals one only at the temporal median. Hence

$$P\tau_i = \mathbf{e}_{m_i}. \tag{S51}$$

Consecutive triangles share one edge, triangles two positions apart share no edge but may have the same median, and more distant triangles have neither contribution. Equations (S41)–(S42) are exactly the matrix entries of $\beta|_Z$. Taking its radical dimension in (S50) proves the theorem. □

S9 Why the forbidden patterns make the grading finite-state

We now show that H-avoidance turns the triangle-interlace graph into a path forest. This is the key structural reason the grading can be recognized by a small automaton.

Use zero-based sites temporarily and write

$$c_i = \mathbf{1}_{\{i \to i+1\}}, \qquad d_i = \mathbf{1}_{\{i \to i+2\}}. \tag{S52}$$

Transitivity of each consecutive triangle is

$$c_i = c_{i+1} \Longrightarrow d_i = c_i. \tag{S53}$$

The interlace graph has vertex set $T(S)$ and edges

$$i \sim i+1 \iff m_i \neq m_{i+1}, \tag{S54}$$

$$i \sim i+2 \iff m_i = m_{i+2}. \tag{S55}$$

There are no longer-range edges.

Lemma S17 (A non-path contains a local triangle)**.** *If $\Omega_D[T(S)]$ is not a disjoint union of paths, then three consecutive triangle indices $i, i+1, i+2$ span a triangle.*

Proof. If $i \sim i+2$, the allowed ranges of the medians force

$$m_i = m_{i+2} = i+2.$$

Then

$$i \sim i+1 \iff m_{i+1} \neq i+2 \iff i+1 \sim i+2.$$

Thus a distance-two edge has either both adjacent edges or neither. In the absence of a local triangle, every vertex has degree at most two. A cycle is also impossible: its leftmost vertex would have neighbors $i+1$ and $i+2$, and the displayed equivalence would complete a triangle. Hence every component is a path. □

A local interlace triangle uses five consecutive selected sites. Enumerating the transitive orientations of the three overlapping spatial triangles gives four seeds:

$c_0c_1c_2c_3$	$d_0d_1d_2$	(m_0, m_1, m_2)
0101	001	$(2, 1, 2)$
0101	011	$(2, 3, 2)$
1010	100	$(2, 3, 2)$
1010	110	$(2, 1, 2)$.

(S56)

Spatial reflection pairs the first with the second and the third with the fourth, so two right-propagating cases suffice.

Let $s_j = \mathbf{1}_{\{j \in S\}}$. Feasibility at a selected site j is

$$s_{j-2}(1 + d_{j-2}) + s_{j-1}(1 + c_{j-1}) + s_{j+1}c_j + s_{j+2}d_j = 0. \tag{S57}$$

For either right-propagating seed, feasibility at site 3 gives $s_5 d_3 = 1$. Thus the selected run extends to site 5 and $d_3 = 1$.

Let $R \geq 5$ be the right endpoint of the maximal consecutive selected run containing sites $0, \ldots, 4$. At an interior site of this run, (S57) becomes

$$d_j + d_{j-2} + c_{j-1} + c_j = 0. \tag{S58}$$

The quantity

$$I_j = c_{j-1} + d_{j-2} + d_{j-1} \tag{S59}$$

is invariant along the run. It equals one for both seeds. Combining this with feasibility at the two final selected sites yields

$$s_{R+2} d_R = 1. \tag{S60}$$

So the run cannot simply stop: site $R+1$ is absent, site $R+2$ is selected, and $R \to R+2$.

The remaining propagation is finite. For the seed $(1010; 100)$, transitivity forces $c_4 = d_4 = 1$, and the translated H_0 has signature $110011d_5$, one of two forbidden signatures. For the seed $(0101; 001)$, runs of lengths $R = 5, 6, 7$ immediately produce a forbidden signature. Longer runs enter one of three states P_n, Q_n, R_n. The exact transition table is

state	c_{n+1}	outcome
P_n	0	1110011 forbidden
P_n	1	violates transitivity
Q_n	0	1111101 forbidden
Q_n	1	P_{n+1}
R_n	0	Q_{n+1}
R_n	1	R_{n+1}.

(S61)

At the right boundary, P_n violates transitivity while Q_n and R_n produce the same two forbidden signatures. The full derivation of the states and their initial conditions appears in section S18; the accompanying verifier checks each finite implication exactly.

Table S1. The local compatibility table. All arithmetic in the last column is over $\mathbb{F}_2$.

R_v	r_v	m_v	type	condition on X
1	0	$x_v y_v$	A	$p_v = 0$
1	1	y_v	B	$p_v + X_v = 1$
0	1	x_v	C	$X_v = 0$
0	0	x_v^2	A	$p_v = 0$
0	0	1	B	$p_v + X_v = 1$

Theorem S18 (Forbidden-H path dichotomy)**.** *Let D be an acyclic orientation of P_M^2 and S a feasible support. If $\Omega_D[T(S)]$ is not a disjoint union of paths, then D contains one of the twenty forbidden orientations. In particular, for an H-avoiding heap,*

$$\Omega_D[T(S)] \text{ is a path forest for every feasible } S. \tag{S62}$$

Proof. By lemma S17, a non-path graph contains a local interlace triangle. Its orientation is one of the four seeds (S56). The preceding propagation forces a forbidden H for the two right-propagating seeds, and spatial reflection handles the other two. □

The binary adjacency matrix of a path on q vertices has nullity one for odd q and zero for even q. Combining theorems S16 and S18 gives the finite-state grading formula.

Corollary S19 (Path grading)**.** *If D is H-avoiding and S is feasible, then*

$$\boxed{\nu_D(S) = c(P_M^2[S]) + o_D(S),} \tag{S63}$$

where $o_D(S)$ is the number of odd-order connected components of the triangle-interlace path forest.

The first term starts a new component exactly when a selected site is preceded by two unselected sites. The second term can be accumulated by retaining only the last two triangle vertices, their connectivity, and the parity of every interlace component that still meets this boundary. After exact minimization, the combined grading transducer has thirty-two reachable states.

S10 Coefficientwise reduction of the Baxter identities

We now reduce the operator identities to two families of signed scalar sums. This step is exact and uses no automata.

S10.1 Local quotient basis

Consider a product of two coefficients $a_D(X)a_D(Y)$. At any site, the degree in x_v, y_v is at most two. Modulo $x_v^2 + y_v^2 = 1$, the degree-two subspace has the basis

$$\mathcal{M} = \{1, x_v, y_v, x_v^2, x_v y_v\}. \tag{S64}$$

Indeed, $y_v^2 = 1 - x_v^2$, and the five displayed functions are independent. Therefore it is enough to prove vanishing separately for every full basis word

$$\boldsymbol{m} = \prod_{v=1}^{M} m_v, \qquad m_v \in \mathcal{M}. \tag{S65}$$

Fix the output graph-Clifford support $R \in \mathbb{F}_2^M$. Write the two supports as

$$Y = X \triangle R = X + R \tag{S66}$$

over $\mathbb{F}_2$. Let

$$p = E_D X, \qquad r = E_D R. \tag{S67}$$

Then the selected outdegree parities of X and Y are p and $p + r$, respectively.

At a site v, compatibility of the pair with feasibility and the chosen basis symbol is exactly one of the five rows in table S1. We call the rows of the first and fourth form type A, the second and fifth type B, and the third type C.

Lemma S20 (Local compatibility)**.** *The rows of table S1 are necessary and sufficient. All other combinations of* (R_v, r_v, m_v) *contribute zero.*

Proof. If $R_v = 1$, exactly one of X_v, Y_v is selected. When $r_v = 0$, both supports see the same outdegree parity, which must be zero for the selected support; the unselected support contributes x_v, giving $x_v y_v$. When $r_v = 1$, feasibility forces the unselected support to have odd parity, so it contributes no x_v; the local symbol is y_v. The condition distinguishing which support is selected is $p_v + X_v = 1$.

If $R_v = 0$, the two supports agree at v. When $r_v = 1$, simultaneous selection is impossible, and the two absent copies have opposite parity, so exactly one contributes x_v. When $r_v = 0$, both absent copies give x_v^2, while simultaneous selection gives $y_v^2 = 1 - x_v^2$; this creates the two final rows. The displayed equations are precisely the feasibility conditions in each case. □

Because p_v depends only on X in the spatial window $v-2, \dots, v+2$, these constraints have radius two. For fixed $(D, R, \boldsymbol{m})$, their solution set is an affine binary space, denoted

$$\mathcal{X}(D, R, \boldsymbol{m}) \subseteq \mathbb{F}_2^M. \tag{S68}$$

S10.2 Commutation and centrality depend only on R

Lemma S21 (Commutation parity)**.** *Let* $X, Y = X + R$ *be feasible. Then*

$$H_D(X)H_D(Y) = (-1)^{e(R)} H_D(Y)H_D(X), \qquad e(R) = |E(P_M^2[R])| \pmod 2. \tag{S69}$$

Proof. The graph-Clifford commutation exponent is $X^{\mathsf{T}} A_M Y = X^{\mathsf{T}} A_M R$. Feasibility implies

$$X^{\mathsf{T}} E_D X = 0, \qquad Y^{\mathsf{T}} E_D Y = 0.$$

Expanding the second identity with $Y = X + R$ gives

$$X^{\mathsf{T}}(E_D + E_D^{\mathsf{T}})R = R^{\mathsf{T}} E_D R.$$

The left side is $X^{\mathsf{T}} A_M R$. The right side counts every edge induced by R once, according to its orientation, and is therefore $e(R)$. □

A graph-Clifford monomial with support R is central exactly when

$$A_M R = 0. \tag{S70}$$

Thus the commutator needs to be checked only for $e(R) = 1$, and inversion only for $A_M R \neq 0$.

S10.3 The two sign characters

Let $\mathcal{T}_A, \mathcal{T}_B, \mathcal{T}_C$ be the sets of sites of the three local types. If Δ is a homogeneous direction of the affine fibre, the table gives

$$(E_D\Delta)_v = 0 \quad (v \in \mathcal{T}_A), \qquad (E_D\Delta)_v = \Delta_v \quad (v \in \mathcal{T}_B), \qquad \Delta_v = 0 \quad (v \in \mathcal{T}_C). \tag{S71}$$

The raw scalar coefficient includes powers of i, the sign from $y_v^2 = 1 - x_v^2$, and the graph-Clifford normal-order sign. Their absolute expression is not needed; only its variation inside the fibre matters.

Lemma S22 (Fibre sign characters)**.** *On every nonempty fibre, after factoring a nonzero scalar independent of* X*, the commutator and inversion signs are respectively*

$$\epsilon_{\mathrm{C}}(X) = (-1)^{|\mathcal{T}_B \cap X|}, \tag{S72}$$

$$\epsilon_{\mathrm{I}}(X) = (-1)^{|\mathcal{T}_A \cap X|}. \tag{S73}$$

Proof. Choose one base point X_0 in the fibre and compare $X_0 + \Delta$ with X_0. A direct normal-order calculation gives, before simplification, a linear character involving the local y^2-reduction signs and $\Delta^{\mathsf{T}} E_D^{\mathsf{T}} R$. Using (S71), every type-A contribution cancels in the commutator character, every type-C coordinate vanishes, and the remaining exponent is

$$\sum_{v \in \mathcal{T}_B} \Delta_v.$$

Hence the ratio of commutator signs is $(-1)^{|\mathcal{T}_B \cap \Delta|}$, which proves (S72) up to the fixed sign at X_0.

The inversion summand contains the additional factor $(-1)^{\nu_D(Y)}$. By lemma S6, translating by Δ changes this factor by $(-1)^{|\Delta|}$. Since Δ vanishes on $\mathcal{T}_C$, multiplying the commutator character by $(-1)^{|\Delta|}$ replaces type B by type A. This proves (S73). □

A fully expanded derivation is included in section S17, and the automaton audit independently compares these signs with direct graph-Clifford multiplication.

S10.4 The scalar coefficient sums

Proposition S23 (Coefficient reduction)**.** *To prove the forward implication in theorem 1, it is enough to prove the following two identities for every H-avoiding heap D, every R, and every basis word $\boldsymbol{m}$:*

$$C_{D,R,\boldsymbol{m}}(z,w) = \sum_{X\in\mathcal{X}(D,R,\boldsymbol{m})} (-1)^{|\mathcal{T}_B\cap X|} z^{\nu_D(X)} w^{\nu_D(X+R)} = 0 \qquad \text{if } e(R)=1, \tag{S74}$$

$$I_{D,R,\boldsymbol{m}}(z) = \sum_{X\in\mathcal{X}(D,R,\boldsymbol{m})} (-1)^{|\mathcal{T}_A\cap X|} z^{\nu_D(X)+\nu_D(X+R)} = 0 \qquad \text{if } A_M R \neq 0. \tag{S75}$$

Proof. Fix R and $\boldsymbol{m}$. By lemma S20, the contributing support pairs are exactly the fibre. If $e(R)=0$, the two graph-Clifford monomials commute and the commutator contribution vanishes before summation. If $e(R)=1$, their difference is twice one order, and lemma S22 gives (S74) up to a fixed nonzero scalar. Similarly, (S70) isolates the noncentral inversion coefficients, and the inversion part of lemma S22 gives (S75). The local basis (S64) is independent, so coefficientwise vanishing proves the identities in $\mathscr{R}_M$. □

S11 Finite weighted automata for all coefficients

The sums (S74)–(S75) are finite-state because every ingredient has bounded spatial memory:

- acyclic orientations are generated by the ternary insertion symbol q;
- H-avoidance is a regular language of fixed forbidden windows;
- the compatibility table has radius two;
- the path grading (S63) is recognized by a finite transducer;
- the two signs, $e(R)$, and noncentrality are parity accumulators.

S11.1 The two component automata

Forbidden-pattern automaton. The orientation stream begins with the first edge orientation and continues with symbols $q\in\{0,1,2\}$. A suffix automaton retains enough orientation bits to decide every translated seven-edge signature when it closes. Deterministic-automaton minimization gives forty-eight accepting states and one dead state. Its transition table is `h_table.hpp`.

Grading transducer. For a support S, the component term $c(P_M^2[S])$ starts a new component exactly when the new bit is one and the preceding two bits are zero. The odd-path term stores the last two triangle vertices, whether they lie in the same open interlace component, and the order parity of every component meeting the boundary. When a component leaves the boundary, the transducer emits one precisely when its order is odd. Exact minimization gives thirty-two reachable states. A transition emits a bit $d\in\{0,1\}$, and the total emitted sum, including a terminal flush, is $\nu_D(S)$. Its table is `grade_table.hpp`.

S11.2 The coefficient automaton

The external alphabet is

$$\Sigma = \{0,1,2\}\times\{0,1\}\times\mathcal{M}, \qquad |\Sigma| = 30. \tag{S76}$$

A symbol (q,r,m) supplies the next orientation insertion, the next bit of R, and the next local quotient-basis symbol. The automaton nondeterministically chooses the next bit of X. The bit of $Y=X+R$ is then fixed.

A raw state is the tuple

$$(h; c_2,c_1,d_2,d_1; X_{[-4]}, R_{[-4]}; m_0,m_1; g_X,g_Y; e_R,n_R). \tag{S77}$$

Here:

- h is the forbidden-pattern DFA state;
- c_2,c_1,d_2,d_1 are the orientation boundary bits;

Algorithm S2 One coefficient-automaton transition

Require: raw state s, input $(q, r_{\mathrm{new}}, m_{\mathrm{new}})$
advance the H-avoidance state with q; reject on the dead state
compute the new orientation bits $(c_{\mathrm{new}}, d_{\mathrm{new}})$
for $x_{\mathrm{new}} \in \{0, 1\}$ **do**
    set $y_{\mathrm{new}} = x_{\mathrm{new}} + r_{\mathrm{new}} \pmod 2$
    form the five-site X- and R-windows centered two sites behind
    evaluate $p_v = (E_D X)_v$ and $r_v = (E_D R)_v$
    determine the local type from table S1; reject if none applies
    advance the two grading transducers and record emissions (d_X, d_Y)
    update e_R and the noncentrality witness
    emit weight $(-1)^{\mathbf{1}_{\{v\in\mathcal{T}_B\}} X_v} z^{d_X} w^{d_Y}$ for commutation
    emit weight $(-1)^{\mathbf{1}_{\{v\in\mathcal{T}_A\}} X_v} z^{d_X + d_Y}$ for inversion
    shift all histories and output the successor state

- $X_{[-4]}, R_{[-4]}$ are four-bit histories, oldest first;
- m_0, m_1 are the two basis symbols whose radius-two neighborhoods are not yet complete;
- g_X, g_Y are the two thirty-two-state grading states;
- e_R is the parity of edges already closed inside R;
- n_R records whether noncentrality $A_M R \neq 0$ has been witnessed.

When a new symbol and branch bit arrive, the site two positions behind is complete. The transition checks its row of table S1, advances both grading states, emits powers of z, w, updates the appropriate sign parity, and shifts the boundary.

The first two sites are handled by $200 = 2 \cdot 2 \cdot 5 \cdot 2 \cdot 5$ external prefixes, each of which sums over the four choices of the initial X-bits. At the right boundary, two zero sites are appended to close the pending local tests and flush the grading transducers. For commutation, a final state is accepted only if $e_R = 1$. For inversion, it is accepted only if the noncentrality witness is present.

S11.3 Reachability and independent semantic audit

Breadth-first closure gives

$$690{,}475 \text{ reachable states}, \qquad 3{,}817{,}800 \text{ transitions}. \tag{S78}$$

The commutator has 5,857 accepting states and 66,394 states that can reach one. Inversion has 12,092 accepting states and 69,874 coaccessible states.

The state encoding was audited independently, and the full audit was rerun for this version. A separate program enumerates every H-avoiding heap through $M = 9$, every external coefficient word encountered by the automaton, and every support branch. It compares:

- local compatibility with direct feasibility of X and $X + R$;
- transducer emissions with direct Gaussian elimination on K_D;
- sign bits with direct graph-Clifford multiplication and y^2-reduction;
- $e(R)$ with direct induced-edge parity;
- the noncentrality flag with $A_M R$.

The audit covers 22,783,498 branches with no discrepancy.

S11.4 Exact formal weighted bisimulation

The raw automaton can be quotiented before choosing numerical values of z, w. For a state s, an input symbol a, and a current partition block B, aggregate all transitions

$$s \xrightarrow{a,\ cz^p w^q} t, \qquad t \in B,$$

Algorithm S3 Exact formal bisimulation quotient

Require: coaccessible weighted automaton $\mathcal{W}$
partition states by their exact final monomial weight
repeat
  for all states s **do**
    $\mathrm{sig}(s) \leftarrow$ final weight of s
    for all input symbols a **do**
      for all outgoing transitions $s \xrightarrow{a, cz^p w^q} t$ **do**
        add c to the entry indexed by $(a, [t], p, q)$
  refine the partition by equality of $\mathrm{sig}(s)$
until the partition is unchanged
aggregate transitions between final blocks and delete zero sums

by the tuple (a, B, p, q), summing the integer coefficients c. Two states are equivalent at the next refinement step when they have the same final weight and all such aggregated transition data agree. Iteration to stability is exact weighted bisimulation.

The resulting exact quotient sizes are

$$n_{\mathrm{C}} = 9{,}756, \qquad n_{\mathrm{I}} = 7{,}195. \tag{S79}$$

The sparse quotients are stored in `comm_formal.txt` and `inv_formal.txt`. Their format is described in section S20.

For either quotient, let M_a be the transition matrix for $a \in \Sigma$, let $u_1, \dots, u_{200}$ be the initial row vectors, and let f be the final column. Every coefficient at every length is exactly

$$u_i M_{a_1} \cdots M_{a_k} f. \tag{S80}$$

This is the standard linear representation of a noncommutative rational series [6].

S12 Exact invariant-module certificates

The final step is a finite linear-algebra lemma.

Lemma S24 (Backward invariant-module criterion)**.** *Let K be a field, $M_a \in K^{n\times n}$, $u_i \in K^{1\times n}$, and $f \in K^{n\times 1}$. Suppose there is a subspace $\mathcal{B} \subseteq K^n$ such that*

$$f \in \mathcal{B}, \qquad M_a \mathcal{B} \subseteq \mathcal{B} \quad (a \in \Sigma), \qquad u_i b = 0 \quad (i = 1, \dots, 200;\ b \in \mathcal{B}). \tag{S81}$$

Then

$$u_i M_{a_1} \cdots M_{a_k} f = 0 \tag{S82}$$

for every word and every initial row.

Proof. By invariance, $M_{a_1} \cdots M_{a_k} f \in \mathcal{B}$. Every u_i annihilates $\mathcal{B}$. □

S12.1 The two certificates

For the commutator quotient, the supplied basis spans

$$\mathcal{B}_{\mathrm{C}} \subseteq \mathbb{Q}(z, w)^{9756}, \qquad \dim \mathcal{B}_{\mathrm{C}} = 590. \tag{S83}$$

Its row-echelon encoding has 72,505 nonzero entries and 1,445 distinct rational functions. The exact verifier checks:

1. every one of the 200 initial rows annihilates every basis vector;
2. the final column reduces to zero modulo the basis;
3. all
$$590 \cdot 30 = 17{,}700 \tag{S84}$$
transition images reduce to zero.

Algorithm S4 Exact symbolic certificate verification

Require: formal quotient, reconstructed row-echelon basis $\{b_p\}$
create the exact field $K = \mathbb{Q}(z, w)$ or $\mathbb{Q}(z)$
parse all sparse transition, initial, and final weights into K
for all basis vectors b_p and initial rows u_i **do**
    verify $u_i b_p = 0$
reduce f by the row-echelon basis; require zero remainder
for all basis vectors b_p and symbols $a \in \Sigma$ **do**
    compute the sparse image $M_a b_p$
    reduce it by the basis pivots in increasing order
    require zero remainder
accept only if every exact equality succeeds

Every operation is performed in the rational-function field $\mathbb{Q}(z, w)$.

For inversion, the certificate is

$$\mathcal{B}_{\mathrm{I}} \subseteq \mathbb{Q}(z)^{7195}, \qquad \dim \mathcal{B}_{\mathrm{I}} = 427. \tag{S85}$$

Its basis has 62,836 nonzero entries and 330 distinct rational functions. The verifier checks all

$$427 \cdot 30 = 12{,}810 \tag{S86}$$

transition images over $\mathbb{Q}(z)$, together with the seed and final conditions.

The sparse bases were discovered by evaluating at generic points over $\mathbb{F}_{65521}$, computing invariant spaces, and reconstructing rational functions from multiple evaluations. This discovery phase is not logically trusted. The verifier reads the reconstructed expressions into a characteristic-zero field and rechecks every condition in (S81). A wrong interpolation, modular collision, or corrupted candidate basis therefore causes an exact verification failure.

S12.2 Vanishing of all coefficients

Apply lemma S24 to the two quotients. Equation (S80) vanishes for every input word and every length. By construction of the automata, this gives

$$C_{D,R,\boldsymbol{m}}(z, w) = 0 \tag{S87}$$

for every H-avoiding D, every anticommuting R, and every local basis word, and

$$I_{D,R,\boldsymbol{m}}(z) = 0 \tag{S88}$$

for every noncentral R. The coefficient sums are polynomials; their embeddings into the rational-function fields are injective, so rational-function vanishing is the desired formal polynomial identity.

We have proved commutation and centrality. Finally,

$$\mathsf{A}_D(z)\mathsf{A}_D(-z) = \mathsf{A}_D(-z)\mathsf{A}_D(z) = \mathsf{A}_D(z)\mathsf{A}_D(-z)|_{z\mapsto -z}\,,$$

so the central product is even in z. It belongs to $\mathcal{Z}_M[z^2]$.

Proof of theorem 1. If D is H-avoiding, theorem S14 gives the degree bound. The commutator certificate proves (S29), and the inversion certificate proves (S30). Thus D has the canonical Baxter property. Conversely, a forbidden copy violates $[Q_{D,1}, Q_{D,3}] = 0$ by corollary S13. □

S13 Analytic checks and special subclasses

The computer-assisted proof is global, but several large subclasses and limiting cases have direct analytic explanations.

S13.1 Stable supports

If S is a stable set of P_M^2, then $N_{D,S} = K_D(S) = 0$, independently of D. Hence

$$\nu_D(S) = |S|.$$

This recovers the ordinary independent-set grading of the original FFD transfer matrix. It also explains why deleting the diagonal of the footprint Gram matrix is essential: keeping the full Gram matrix would assign degree zero to every stable support.

S13.2 The three regular circuit families

The half-words in (19) have periodic local orientations. In the edge order $AB, BC, BD, CD, CE, DE, EF$, their translated six-site signatures are

Family	Signatures, starting with $a = 1$
I	`1111111`
II	`1011101`, `1110111`
III	`0110001`, `1000110`, `1011101`

The lists repeat with spatial periods one, two, and three. None is forbidden, proving acceptance at every compatible length. This check uses the same conventions as the scalar spectral constructions in Ref. [4]; no identification with a different transfer polynomial is required in the proof above.

S14 A constructive sector-aware inversion polynomial

The classification theorem proves *central* inversion. This section gives an explicit scalar polynomial in every central sector, without an additional assumption that all sectors share the same polynomial. It also specifies the finite-frontier algorithm implemented in `spectrum.py`.

S14.1 The center has bounded dimension

Let A_M be the binary adjacency matrix of P_M^2. The monomial $h^R = h_1^{R_1} \cdots h_M^{R_M}$ is central exactly when $A_M R = 0$. With $R_j = 0$ outside $[M]$, its components satisfy

$$R_{j+2} = R_{j-2} + R_{j-1} + R_{j+1} \pmod 2. \tag{S89}$$

The first boundary equation is $R_3 = R_2$. Thus R_1, R_2 determine the entire vector; the right boundary can only impose further constraints. Consequently $d := \dim\ker A_M \le 2$ and there are at most four central support masks. The case $M = 1$ is immediate. Binary elimination gives both a kernel basis and the irreducible dimension $2^{(M-d)/2}$.

For a convenient real-coefficient bookkeeping convention, put

$$g_S = i^{|S|} h^S, \qquad b_D(S) = (-1)^{\iota_D(S)} \prod_{j\in S} y_j \prod_{\substack{j\notin S \\ e_D(j,S)=0}} x_j, \tag{S90}$$

where $\iota_D(S)$ counts reversed spatial edges in the selected temporal word. Then

$$\mathsf{A}_D(z) = \sum_{S\in\mathcal{F}(D)} b_D(S) z^{\nu_D(S)} g_S, \tag{S91}$$

and multiplication is

$$g_S g_T = \mu(S,T) g_{S\triangle T}, \qquad \mu(S,T) = (-1)^{|S\cap T| + \sum_{i>j} S_i T_j (A_M)_{ij}}. \tag{S92}$$

The first exponent comes from the powers of i, and the second from restoring spatial order.

S14.2 Exact coefficient formula

For every central support R, define

$$p_R(z) = \sum_{\substack{S,T\in\mathcal{F}(D) \\ T = S\triangle R}} b_D(S) b_D(T) \mu(S,T) (-1)^{\nu_D(T)} z^{\nu_D(S)+\nu_D(T)}. \tag{S93}$$

The classification theorem gives the operator identity

$$\mathsf{A}_D(z)\mathsf{A}_D(-z) = \sum_{R\in\ker A_M} p_R(z) g_R. \tag{S94}$$

Consequently the required scalar polynomial is

$$\boxed{p_{D,\chi}(z) = \sum_{R \in \ker A_M} \chi(g_R) p_R(z).} \tag{S95}$$

This is a finite, explicit prescription for arbitrary admissible heaps. It requires no physical matrix representation.

To enumerate the central characters unambiguously, take a binary kernel basis $R^{(1)}, \dots, R^{(d)}$ and let

$$p_\ell = |R^{(\ell)}| + |E(F_M[R^{(\ell)}])| \pmod 2, \qquad C_\ell = i^{p_\ell} g_{R^{(\ell)}}.$$

The C_ℓ are commuting Hermitian involutions. Choose $\chi(C_\ell) = \zeta_\ell \in \{\pm 1\}$ independently, set $\chi(g_{R^{(\ell)}}) = i^{-p_\ell}\zeta_\ell$, and use (S92) for every other radical mask. This avoids treating non-Hermitian central monomials as though their characters had to be real signs.

For $R = 0$, feasibility implies $|E(F_M[S])|$ is even and $\nu_D(S) \equiv |S| \pmod 2$. Therefore (S93) reduces to

$$p_0(z) = \sum_{S \in \mathcal{F}(D)} z^{2\nu_D(S)} \prod_{j \in S} y_j^2 \prod_{\substack{j \notin S \\ e_D(j,S)=0}} x_j^2. \tag{S96}$$

In the faithful Pauli representation this is the normalized *full* trace of the inversion product. A sector trace can also retain terms with $R \neq 0$. Thus (S96) alone is a sector polynomial only after scalar inversion in the full algebra has been proved or checked. No such extra claim is needed for (S95).

S14.3 Bounded-frontier evaluation

Fix R and scan the sites from left to right. At each site choose the bit S_j; then $T_j = S_j + R_j$ is fixed. Two copies of the 32-state grading transducer accumulate $\nu_D(S)$ and $\nu_D(T)$. Two additional past bits of S suffice for the radius-two feasibility and angle factors; the corresponding past bits of T follow from R. There are at most $4 \cdot 32^2 = 4096$ frontier states. Each stores a polynomial of degree at most $2s_M$.

Let $D_{jk} = 1$ when the spatial edge $j - k$ is directed from j to k. When site j is finalized, the local angle factor is $w_j(S)w_j(T)$, where

$$w_j(S) = \begin{cases} y_j, & S_j = 1, \\ x_j, & S_j = 0,\ e_D(j,S) = 0, \\ 1, & S_j = 0,\ e_D(j,S) = 1. \end{cases}$$

Reject selected sites with odd outdegree parity in either support. Using $\nu_D(T) \equiv |T|$, the sign emitted at that site is $(-1)^{\eta_j}$ with

$$\eta_j = S_j T_j + T_j + \sum_{k=\max(1,j-2)}^{j-1} [D_{jk}(S_k S_j + T_k T_j) + S_j T_k] \pmod 2. \tag{S97}$$

The first two terms account for the g-basis product and inversion parity; the sum accounts for the two temporal reorderings and the spatial product sign. Every spatial edge appears exactly once.

Algorithm S5 Sector-polynomial computation without many-body matrices

Require: heap D, angle data x_j, y_j, grading transducer
1: reject if a translated forbidden signature occurs
2: find the at most four masks $R \in \ker A_M$ by binary elimination
3: **for all** such R **do**
4: initialize the four choices for the first two bits of S; set $T = S + R$
5: initialize the two grade states and their emitted degrees
6: **for** $j = 1, \dots, M$ **do**
7: **for all** current frontier states and choices of the new bit S_{j+2} **do**
8: set $T_{j+2} = S_{j+2} + R_{j+2}$; outside the chain use zero bits
9: finalize both feasibility tests and the two angle factors at site j
10: multiply the stored polynomial by $(-1)^{\eta_j} w_j(S)w_j(T)$
11: advance the two grade states and shift the polynomial by their total emission
12: merge identical successor states by addition
13: flush both grade states and sum terminal weights to obtain $p_R(z)$
14: evaluate (S95) for each central character
15: solve the resulting scalar root polynomials; retain sector labels

At initialization the degree already emitted by two bits (s_1, s_2) is $s_1 \vee s_2$; add the analogous contribution for T. Two zero-padding steps close the right boundary. The supplied implementation follows these conventions exactly. Polynomial coefficient vectors have $2s_M + 1$ entries, so the arithmetic count is $O(Ms_M)$ and the working storage is $O(s_M)$, with fixed constants from the state bound. These are arithmetic-operation bounds, not bit-complexity or numerical-conditioning guarantees. The reference implementation uses floating arithmetic; its direct independent audit uses rational angle data for the underlying algebraic checks.

S15 Root selection, sign constraints, and physical quasienergies

S15.1 Selecting one root from each pair

For generic parameters let

$$p_{D,\chi}(z) = a_0^2 \prod_{k=1}^{s_M} \left(1 - \frac{z^2}{r_{\chi,k}^2}\right), \qquad a_0 \neq 0, \tag{S98}$$

with distinct nonzero pairs. Any common transfer eigenvalue $\Lambda(z)$ satisfies $\Lambda(0) = a_0$ and $\Lambda(z)\Lambda(-z) = p_{D,\chi}(z)$. It must contain exactly one zero from each pair: neither omission nor double selection is compatible with a simple zero of $p_{D,\chi}$. Thus

$$\Lambda_{\chi,\sigma}(z) = a_0 \prod_{k=1}^{s_M} \left(1 + \frac{\sigma_k z}{r_{\chi,k}}\right). \tag{S99}$$

This argument identifies the possible polynomials; completeness and multiplicities follow from the stable-set specialization in Theorem S11.

S15.2 The sign code is explicit for the open FFD chain

Let $I = \{1, 4, 7, \ldots\} \cap [M]$, with $|I| = s_M$, and consider the commuting generators h_j on I. A vector supported on I lies in $\ker A_M$ only if its successive selected bits are all equal. The right boundary allows the nonzero constant vector precisely when $M \equiv 1 \pmod 3$. Hence

$$\dim\left(\mathbb{F}_2^I \cap \ker A_M\right) = \delta_M := \begin{cases} 1, & M \equiv 1 \pmod 3, \\ 0, & \text{otherwise.} \end{cases} \tag{S100}$$

At the decoupled specialization, the sector therefore contains all 2^{s_M} sign choices when $\delta_M = 0$, and one parity half of them when $\delta_M = 1$. Continuing the simple root pairs can only relabel coordinates or reverse their signs. The code remains either the full cube or a single product constraint; no more general binary code is needed for this chain.

When $\delta_M = 1$, the highest charge Q_{D,s_M} is central: its eigenvalue is the same for all allowed signs in each sector, and the identity extends polynomially from the generic locus. Write $q_{\chi,s_M} = \chi(Q_{D,s_M})$. Comparing leading coefficients in (S99) fixes the sign constraint directly,

$$\boxed{\prod_{k=1}^{s_M} \sigma_k = \frac{q_{\chi,s_M}}{a_0} \prod_{k=1}^{s_M} r_{\chi,k} \ \in \{\pm 1\}.} \tag{S101}$$

Its value depends on the chosen representatives of the root pairs. It is not legitimate to choose that parity independently of the sector.

The number of distinct generic transfer eigenpolynomials in one irreducible sector is $2^{s_M - \delta_M}$. Each has multiplicity

$$2^{(M-d)/2 - s_M + \delta_M}. \tag{S102}$$

A physical representation may contain additional copies of an irreducible sector. Special angles can merge distinct Floquet eigenvalues even when the transfer eigenpolynomials remain distinguishable.

S15.3 Reality of the physical root pairs

For real angles, every transfer eigenpolynomial obeys

$$\overline{\Lambda(\bar{z})} = \Lambda(-z). \tag{S103}$$

Suppose a root pair were not purely imaginary. Complex conjugation combined with $z \mapsto -z$ would either exchange the two members of that same pair, which contradicts one-root selection, or map it to a different pair. In the latter case the signs on those two pairs would have to be fixed multiples of one another in *every* allowed sign vector, by (S103). The full sign cube has no such relation. A single parity code also has no such two-coordinate relation when $s_M \geq 3$. Thus all pairs are purely imaginary in these cases.

There is only one exceptional small code to check: $M = 4$, $s_M = 2$, where the two signs in a fixed sector are correlated. Direct exact quotient-ring multiplication for all eighteen four-site heaps proves scalar inversion in the full algebra. The two sectors of the central product $h_1 h_4$ then supply opposite parity halves of the *same* root polynomial, giving the full two-coordinate cube. The preceding argument applies to their union. The archive includes the all-angle four-site calculation (and checks at $M = 2, 3, 5$); it does not rely on a sampled-angle discriminant. When $s_M = 1$, (S103) already makes the single nonzero root imaginary.

Therefore, on the generic real-angle locus, write $r_{\chi,k} = iu_{\chi,k}$, $u_{\chi,k} > 0$. The normalized polynomial

$$P_{D,\chi}(x) = \frac{p_{D,\chi}(i\sqrt{x})}{a_0^2} = \prod_k \left(1 - \frac{x}{u_{\chi,k}^2}\right) \tag{S104}$$

has positive roots. Unitarity at $z = 1$ gives $p_{D,\chi}(1) = 1$ and hence

$$a_0^2 \prod_k (1 + u_{\chi,k}^{-2}) = 1. \tag{S105}$$

Substituting into (S99) yields the phase form

$$v_{\chi,\sigma} = \operatorname{sgn}(a_0) \exp\left[-i \sum_k \sigma_k \arctan(u_{\chi,k}^{-1})\right]. \tag{S106}$$

This is the desired polynomial-roots-to-quasienergies solution, including the sector restriction rather than silently doubling its spectrum.

S15.4 Nongeneric points and numerical use

The operator identities and unnormalized inversion polynomials hold for every angle. The factorization formulas using a_0^{-1}, simple roots, or distinct occupations require the generic assumptions explicitly stated above. At vanishing cosines or coalescing roots, take limits of the unnormalized identities. The supplied numerical solver does not discard small leading coefficients by an absolute trimming rule; near singular points or strongly separated root scales, higher-precision polynomial arithmetic and certified root isolation are preferable. A floating root computation is not an exact certificate for a spectrum.

S16 The twenty forbidden signatures

Order the seven edges of H as

$$(AB, BC, BD, CD, CE, DE, EF),$$

and write a one when the first endpoint precedes the second. The twenty forbidden signatures are

0000010, 0001100, 0011000, 0011001, 0011101,
0100000, 0100001, 0100011, 0110011, 0111101,
1000010, 1001100, 1011100, 1011110, 1011111,
1100010, 1100110, 1100111, 1110011, 1111101.

They are generated from the five words in (S24) by temporal reversal and spatial reflection. The exact local verifier constructs this set independently and compares it with the nonzero commutator cases.

S17 Detailed sign derivation

We record a direct derivation of lemma S22. Fix $(D, R, \boldsymbol{m})$, let $Y = X + R$, and use the spatial normal order $h^S = h_1^{S_1} \cdots h_M^{S_M}$.

For a feasible support S, write

$$H_D(S) = \eta_D(S) h^S, \qquad \eta_D(S) \in \{\pm 1\}.$$

The product sign is

$$H_D(X) H_D(Y) = \eta_D(X)\eta_D(Y)(-1)^{\sum_{i>j} X_i Y_j A_{ij}} h^R. \tag{S107}$$

All powers of i, all symbols fixed by $(R, r, \boldsymbol{m})$, and the sign at one base point may be factored out. The remaining dependence on X is a binary affine character. One convenient representative is

$$\chi(X) = X^{\mathsf{T}} E_D^{\mathsf{T}} R + \mathbf{1}_Z^{\mathsf{T}} X, \tag{S108}$$

where Z is the subset of type-B sites whose chosen local basis symbol is 1. The first term combines temporal and spatial normal-order signs; the second includes the local basis reduction.

Let Δ satisfy the homogeneous equations (S71). Since $r = E_D R$,

$$\begin{aligned} \chi(X+\Delta) - \chi(X) &= \Delta^{\mathsf{T}} E_D^{\mathsf{T}} R + \mathbf{1}_Z^{\mathsf{T}} \Delta \\ &= \sum_v (E_D \Delta)_v R_v + \sum_{v \in Z} \Delta_v. \end{aligned} \tag{S109}$$

At a type-A site, $(E_D\Delta)_v = 0$. At a type-B site, $(E_D\Delta)_v = \Delta_v$. Type C has $R_v = 0$ and $\Delta_v = 0$. The first sum therefore counts type-B sites with $R_v = 1$, while Z counts the type-B sites with $R_v = 0$. Together they give

$$\chi(X+\Delta) - \chi(X) = \sum_{v \in \mathcal{T}_B} \Delta_v.$$

Thus the commutator sign is a fixed constant times $(-1)^{|\mathcal{T}_B \cap X|}$.

For inversion, the summand has the additional factor $(-1)^{\nu_D(Y)}$. Since $\nu_D(Y) \equiv |Y| \pmod 2$, its ratio under $X \mapsto X + \Delta$ is $(-1)^{|\Delta|}$. Because Δ vanishes on type C,

$$|\mathcal{T}_B \cap \Delta| + |\Delta| = |\mathcal{T}_A \cap \Delta| \pmod 2.$$

This gives the inversion character.

S18 Full propagation table for the path theorem

For completeness, we spell out the finite portion of the proof summarized in section S9. Consider the right-propagating seed

$$c_0 c_1 c_2 c_3 = 0101, \qquad d_0 d_1 d_2 = 001, \qquad d_3 = 1.$$

The terminal runs $R = 5, 6, 7$ yield

R	translated copy	forbidden signature
5	H_0	0011101
6	H_1	0110011 or 0111101
7	H_2	1110011 or 1111101

For $R \geq 8$, define for $n \geq 6$

$$\begin{aligned} P_n: &\quad c = 0101\,1^{n-5}01, && d = 001\,1^{n-4}00, \\ Q_n: &\quad c = 0101\,1^{n-4}0, && d = 001\,1^{n-3}0, \\ R_n: &\quad c = 0101\,1^{n-3}, && d = 001\,1^{n-2}. \end{aligned}$$

Feasibility at $j = 4, 5, 6$, transitivity, and avoidance of H_0, H_1 leave exactly P_6, Q_6, R_6. An interior extension satisfies

$$d_{n+1} = d_{n-1} + c_n + c_{n+1}.$$

The complete transition table is (S61). At the terminal stage $n = R - 2$, the boundary equation fixes

$$c_{n+1} = d_{n-1} + c_n.$$

State P_n violates transitivity; Q_n produces 1110011; and R_n produces 1111101. The second right-propagating seed forces 1100110 or 1100111 after one step. Spatial reflection completes the four cases.

The verifier `verify_h_triangle_path.py` checks the following finite claims independently:

Algorithm S6 Raw grade-transducer update

Require: last two support bits, interlace frontier, new orientation bits and support bit
add 1 to the component count if the new support bit is 1 and the last two bits are 0
determine whether the newly completed consecutive triangle belongs to $T(S)$
if a new triangle vertex is present **then**
    compute its edges to the previous two triangle vertices from their temporal medians
    create or merge frontier components accordingly
    toggle the order parity of the component containing the new vertex
for all frontier components that disappear **do**
    emit 1 if the stored order parity is odd, otherwise emit 0
shift the support and triangle frontiers

1. the list of twenty forbidden signatures;
2. the enumeration of the four local seeds;
3. the feasibility invariant (S59) and terminal jump (S60);
4. all short-run signatures;
5. the six initial length-seven prefixes and the surviving states P_6, Q_6, R_6;
6. every entry of (S61);
7. the three terminal outcomes.

S19 The grading transducer

This appendix describes the finite state used to evaluate (S63). Retain the last two support bits to detect new components of $P_M^2[S]$. For the triangle-interlace contribution, the boundary contains at most the last two triangle vertices. A raw interlace state records:

- whether each of the two vertices is present;
- if both are present, whether they belong to the same component;
- the order parity of every component that still meets the boundary.

Up to relabeling open components, the eleven possibilities are

$$\begin{aligned}&(-,-),\\&(-,0)_0,\ (-,0)_1,\quad (0,-)_0,\ (0,-)_1,\\&(0,0)_0,\ (0,0)_1,\\&(0,1)_{00},\ (0,1)_{01},\ (0,1)_{10},\ (0,1)_{11}.\end{aligned}$$

When a new triangle vertex is introduced, its edges to the previous two vertices are known from the median rules. The update forms the corresponding union of boundary components and toggles the order parity of the component receiving the new vertex. Whenever a component no longer meets the boundary, the transducer emits its parity. The final flush emits the parities of all remaining components.

Combining these eleven states with the previous two support bits gives at most forty-four raw states. Removing unreachable states and minimizing the deterministic output transducer gives the thirty-two states stored in `grade_table.hpp`.

S20 Formal quotient file format

The formal quotient files are sparse ASCII files. The first line contains

$$n \quad n_{\mathrm{seed}} \quad |\Sigma|.$$

The remaining lines have one of three forms:

$$\begin{aligned} \texttt{F i ok s p q}: &\quad f_i = \texttt{ok}\,(-1)^s z^p w^q, \\ \texttt{E src a dst p q c}: &\quad (M_a)_{\texttt{src,dst}} \mathrel{+}= c z^p w^q, \\ \texttt{S i dst p q c}: &\quad (u_i)_{\texttt{dst}} \mathrel{+}= c z^p w^q. \end{aligned}$$

For inversion, every q is zero. Integer coefficients with the same source, symbol, destination, and exponent pair have already been aggregated; zero totals are omitted.

S21 Certificate reconstruction and exact verification

The discovery phase evaluates the formal quotient at several generic points over $\mathbb{F}_{65521}$, computes a small backward invariant space, and records a stable pivot pattern. Coordinate functions are reconstructed as sparse rational functions. The commutator data contain 1,445 distinct functions, while inversion contains 330.

The exact verifier performs the following checks.

1. It constructs $\mathbb{Q}(z, w)$ or $\mathbb{Q}(z)$ using SymPy's exact polynomial domains.
2. It reconstructs every basis entry and checks row-echelon normalization: pivot entries equal one and no entry lies left of its pivot.
3. It parses the formal quotient without numerical specialization.
4. For each basis vector and each seed, it computes the exact dot product and requires zero.
5. It reduces the exact final column by the basis and requires zero remainder.
6. For every basis vector b and symbol a, it forms $M_a b$, performs sparse row reduction, and requires zero remainder.

The computations can be split into arbitrary index ranges. For this version, both complete ranges were independently rerun in a single pass, checking all 17,700 commutator images and all 12,810 inversion images.

S22 Pseudocode for the complete proof pipeline

Algorithm S7 End-to-end certificate generation and verification

```
build and minimize the forbidden-pattern DFA
derive and minimize the grading transducer
breadth-first enumerate reachable coefficient-automaton states
for problem in {commutator, inversion} do
    mark final states satisfying odd induced-edge parity or noncentrality
    remove states that cannot reach a marked final state
    compute the exact formal weighted-bisimulation quotient
    export sparse transition matrices, initial rows, and final column
    evaluate the quotient over a finite field at generic spectral points
    discover a small backward invariant module and stable pivot pattern
    interpolate candidate rational-function basis entries
    reconstruct the basis over characteristic zero
    verify seed annihilation, terminal membership, and all transition images exactly
run independent local, finite-size, and branch-semantic audits
conclude all coefficient series vanish by lemma S24
```

S23 Verification record and compact reproducibility bundle

The accompanying `ffd_certificate.pyz` is a standard Python ZIP application containing the complete source files and certificate data. Its contents can be inspected with any ZIP reader or extracted with `--extractDIRECTORY`; it is not an opaque executable. The outer bundle contains only three files: this single LaTeX source, its PDF, and the code archive. A build and usage guide is included inside the archive. All figures and references are embedded in the LaTeX source.

S23.1 Exact checks rerun for this version

The following computations were completed in the preparation environment, rather than inferred from the earlier release's test record:

Check	Result
Six-site obstruction	All 72 orientations enumerated; exactly the stated 20 have the nonzero full-support $[Q_1, Q_3]$ coefficient.
Sink-nullity closure	908 reachable states; 191 right-flushable states; no nonzero kernel vector vanishing at all sinks.
Triangle and grading reductions	All finite local propagation tables passed; direct triangle-nullity audit through $M = 9$ and grading-table audit through $M = 8$.
Raw coefficient automaton	690,475 states and 3,817,800 transitions regenerated.
Formal quotient regeneration	Both quotient files regenerated and compared byte for byte with the supplied certificates.
Commutator module	All 17,700 transition images, all seed products, and terminal membership checked over $\mathbb{Q}(z, w)$.
Inversion module	All 12,810 transition images, all seed products, and terminal membership checked over $\mathbb{Q}(z)$.
Independent semantic audit	Every admissible heap through $M = 9$; 22,783,498 coefficient branches checked against direct definitions.
Exact angle-ring audit	Every admissible heap at $M = 10$; commutation and central inversion verified coefficientwise.
Small-size scalar audit	Scalar inversion checked in the full angle quotient ring for every heap at $M = 2, 3, 4, 5$.

Finite fields remain discovery tools only. The final module identities are exact equalities in characteristic zero. Independent finite-size checks audit the encodings; they do not replace the invariant-module argument that proves the all-length theorem.

S23.2 Independent checks of the constructive spectral route

The new `audit_new.py` does not import the release's support-expansion or nullity implementations. It constructs predecessor footprints directly and uses its own binary row reduction. For deterministic rational sine/cosine data, it compares the support expansion with a separate outside-in palindrome calculation and verifies the noncentral coefficients of the inversion product exactly. The same checks verify every fixed-support grading, the centrality of the highest charge when $M \equiv 1 \pmod 3$, and the implemented sign constraints. Across 50 admissible permutations at $1 \le M \le 10$, the paired-frontier polynomial calculation agrees with the independent support-pair sum to at most 2.3×10^{-16} in the floating comparison.

For the irregular nine-site example in the letter, it separately constructs $h_j = Z_{j-2}Z_{j-1}X_j$ on nine qubits (omitting factors outside the chain), multiplies the physical gates, and diagonalizes $V = GG^T$. The eight root-predicted eigenvalues each have multiplicity 64; the largest distance of an exact-matrix eigenvalue from the predicted set is 1.2×10^{-14}. This last comparison uses numerical linear algebra; it is a validation of the spectral implementation, not part of the symbolic theorem.

S23.3 Commands

From the directory containing the code archive:

```
python ffd_certificate.pyz --quick
python ffd_certificate.pyz --full
python ffd_certificate.pyz --audit
python ffd_certificate.pyz --extract extracted_sources
python ffd_certificate.pyz --spectrum --example
```

The exact proof uses Python, SymPy, and a C++20 compiler. The numerical solver and independent dense audit additionally use NumPy. The archive contains the verification logs and machine-readable spectral audit. The archived README records the tested versions and gives the single-source PDF build command.